



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO5 AT2 TECHNICAL PRESENTATION

CSA 1707 ARTIFICIAL INTELLIGENCE

Reactive Agent-Based Traffic Control System

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INTRODUCTION

- Traditional traffic signals operate using fixed time intervals.
- They cannot respond effectively to sudden changes in traffic density.
- A Reactive Agent continuously observes the traffic environment.
- It immediately responds to real-time traffic conditions.
- The objective is to reduce congestion, waiting time, and unnecessary signal delays.



Problem Statement

- Traditional traffic signal systems use fixed timings and cannot respond effectively to real-time changes in traffic density. This leads to traffic congestion, increased waiting time, fuel consumption, and inefficient road utilization. Therefore, a Reactive Agent-Based Traffic Control System is required to continuously process real-time traffic data and dynamically adjust signal timings to minimize congestion and improve traffic flow.

Reactive Agent

- A Reactive Agent is an AI agent that:
- Perceives the current environment.
- Processes incoming information.
- Applies predefined condition-action rules.
- Performs an immediate action.
- Does not require long-term planning or memory.

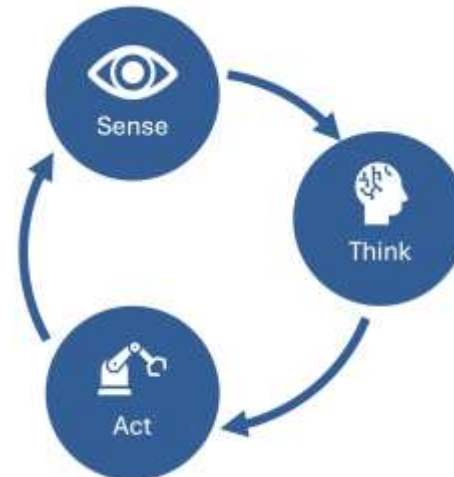
Example

IF traffic density is HIGH
→ **THEN** increase green-light duration.

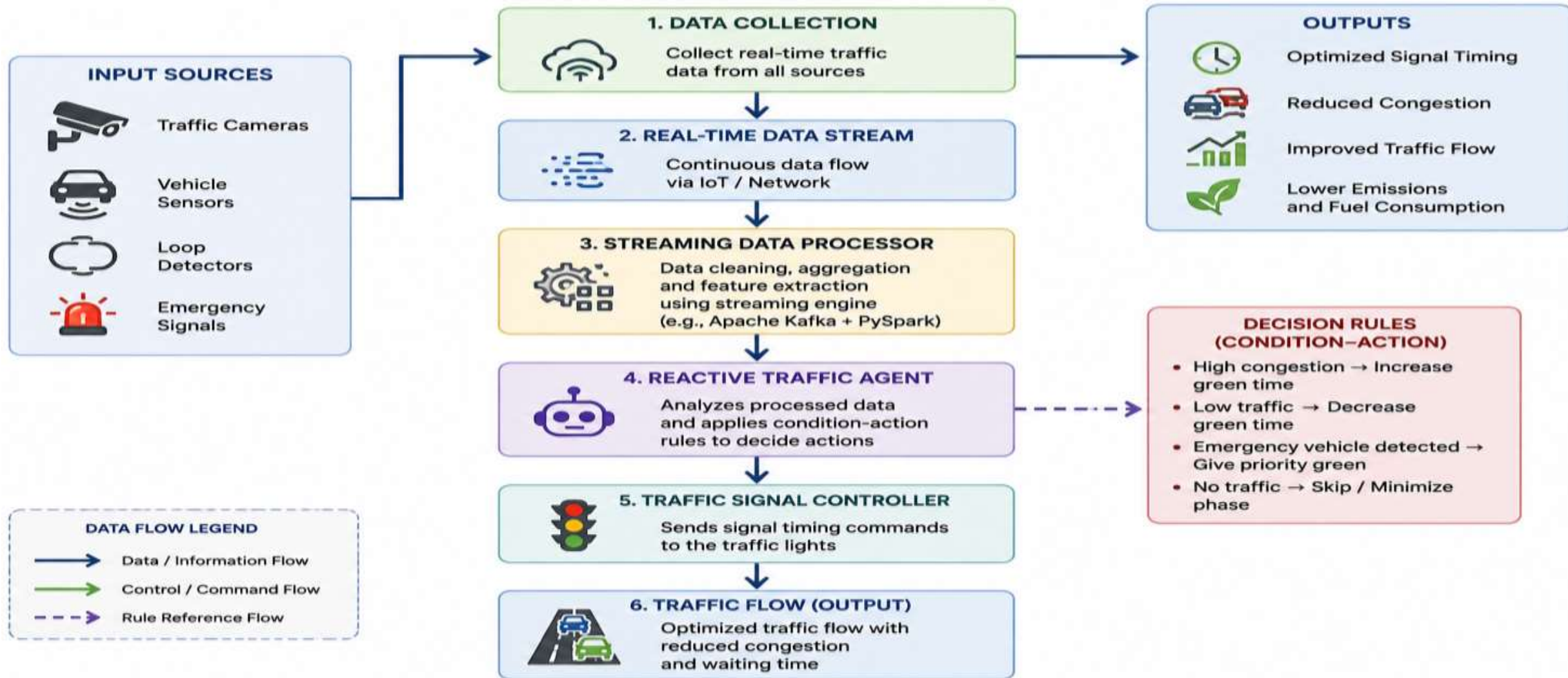
IF traffic density is LOW
→ **THEN** reduce green-light duration.

IF emergency vehicle detected

→ **TH** change green-light duration



System Architecture



Real-Time Data Processing

Streaming Traffic Data

- The system continuously receives traffic information from sensors and cameras.

Pipeline Processing:

**Data Collection → Data Cleaning → Real-Time Analysis → Agent Decision
→ Signal Update**

Example

If the sensor detects:

Lane A → 80 vehicles

Lane B → 25 vehicles

Lane C → 15 vehicles

Lane D → 10 vehicles

The agent identifies **Lane A** as **highly congested** and dynamically increases its green-light duration.

Dynamic Signal Management

- Agent continuously monitors traffic density.
- Adjusts **green-light duration** based on traffic conditions.
- High traffic → **Increase green time.**
- Low traffic → **Decrease green time.**
- Helps reduce **queue length and waiting time.**
- **Example:**
 - Normal traffic → 30 sec green
 - Heavy traffic → 50 sec green

Congestion & Emergency Management

- Reduces congestion by giving priority to heavily crowded lanes.
- Minimizes average vehicle waiting time.
- Detects emergency vehicles in real time.
- Emergency vehicle detected → **Priority green signal.**
- After the vehicle passes, the system returns to normal operation.



Conclusion

- A Reactive Agent provides fast and automatic traffic control.
- Real-time streaming data allows continuous monitoring of traffic conditions.
- Condition-action rules enable immediate signal changes.
- Dynamic signal management helps reduce congestion and waiting time.
- The system can serve as a foundation for AI-powered smart city traffic management.





Thank You